```
1727  0  INSERT FILE='E:/Documents/GitHub/biostatistics-portfolio/_spss_tmp/contraceptive
s-one-way-anova.sps' ERROR=CONTINUE.
1727  0  * Set once: the folder that contains this repository, then run.
1728  0  CD 'E:/Documents/GitHub/biostatistics-portfolio'.
1729  0  GET DATA
1730  0    /TYPE=XLSX
1731  0    /FILE='data/raw/one_Way_ANOVA classwork_2022.xlsx'
1732  0    /SHEET=name 'Sheet2'
1733  0    /READNAMES=on.
1734  0  DATASET NAME DataSet1 WINDOW=FRONT.
```

Dataset Name

Notes

Output Created		03-NOV-2026 15:37:24
Comments		
Input	Filter	<none>
	Weight	<none>
	Split File	<none>
Syntax		DATASET NAME DataSet1 ...
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

Warnings

The active dataset will replace the existing dataset named DataSet1.

```
1735  0  * Encoding: UTF-8.
1736  0
1737  0  *------------------------------------------------------------------------------
----------------------------------------
           ------------------------------------------------------------------------------
---
1738  0  *ONE WAY ANOVA SPSS.
1739  0  *- Analyze - compare means - one way ANOVA.
1740  0  *- Click on dependent and put in dependent list.
1741  0  *- Click independent and put in box labeled factor.
1742  0  *- Click options - descriptives - homogeneity of variances and means plots.
1743  0  *- Click on Post hoc & click Tukey - continue - OK.
1744  0  *------------------------------------------------------------------------------
----------------------------------------
           ------------------------------------------------------------------------------
---
1745  0
```

```
1746  0
1747  0  ONEWAY wtgain BY contraceptive
1748  0    /STATISTICS DESCRIPTIVES HOMOGENEITY
1749  0    /MISSING ANALYSIS
1750  0    /CRITERIA=CILEVEL(0.95).
```

Oneway

Notes

Output Created		03-NOV-2026 15:37:25
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	24
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each analysis are based on cases with no missing data for any variable in the analysis.
Syntax		ONEWAY wtgain BY contraceptive /STATISTICS DESCRIPTIVES HOMOGENEITY /MISSING ANALYSIS /CRITERIA=CILEVEL (0.95).
Resources	Processor Time	00:00:00.03
	Elapsed Time	00:00:00.07

Descriptives

wtgain

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum
					Lower Bound	Upper Bound	
1	6	11.000	1.4711	.6006	9.456	12.544	9.5
2	6	8.000	2.7350	1.1165	5.130	10.870	4.9
3	6	9.000	2.7004	1.1024	6.166	11.834	5.7
4	6	6.000	1.7900	.7308	4.122	7.878	3.8
Total	24	8.500	2.7874	.5690	7.323	9.677	3.8

Descriptives

wtgain

	Maximum
1	13.0
2	11.7
3	12.4
4	8.2
Total	13.0

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2	Sig.
wtgain	Based on Mean	1.315	3	20	.297
	Based on Median	.832	3	20	.492
	Based on Median and with adjusted df	.832	3	12.142	.501
	Based on trimmed mean	1.308	3	20	.299

ANOVA

wtgain

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	78.000	3	26.000	5.164	.008
Within Groups	100.700	20	5.035		
Total	178.700	23			

```
1751  0  *-------------------------------------------------------------------------------
---------------------------------------
           ------------------------------------------------------------------------------
---
1752  0  *From above OUTPUT; note that to check for homoegenity of variances; SPSS perfor
ms a levenes tests with the following hypot
           heses;.
1753  0  *Ho; There is no difference in population variance of the four groups.
1754  0  *Ha: there is a difference in population variance.
1755  0
1756  0  *NOTE; For assumption to be fulfilled the p-value SHOULD NOT BE significant.
1757  0  *-------------------------------------------------------------------------------
---------------------------------------
           ------------------------------------------------------------------------------
---
1758  0  * You can Also see the ANOVA table; we use it to conclude on the hypotheses of i
nterest from our research question.
1759  0  *QUESTION;
1760  0       *Family health worker receives complaints of weight gain from clients on co
ntraceptives.
1761  0      *He has also noticed & is aware that wt gain varies from individual to indiv
idual.
1762  0      *He wants to find out if it is due to genetic variation or contraceptives. H
e compares four different contraceptives.
1763  0
1764  0  *HYPOTHESIS
1765  0  *Ho: there is no difference in weight gain among populations using different con
traceptives; μ1=μ2=μ3=μ4.
1766  0  *Ha: the average weight gain in the population of at least one group is differen
t; μiμj.
1767  0
1768  0  *-------------------------------------------------------------------------------
---------------------------------------
           ------------------------------------------------------------------------------
---
1769  0
1770  0
1771  0
1772  0
1773  0  ONEWAY wtgain BY contraceptive
1774  0    /PLOT MEANS
1775  0    /MISSING ANALYSIS
1776  0    /CRITERIA=CILEVEL(0.95).
```

Oneway

Notes

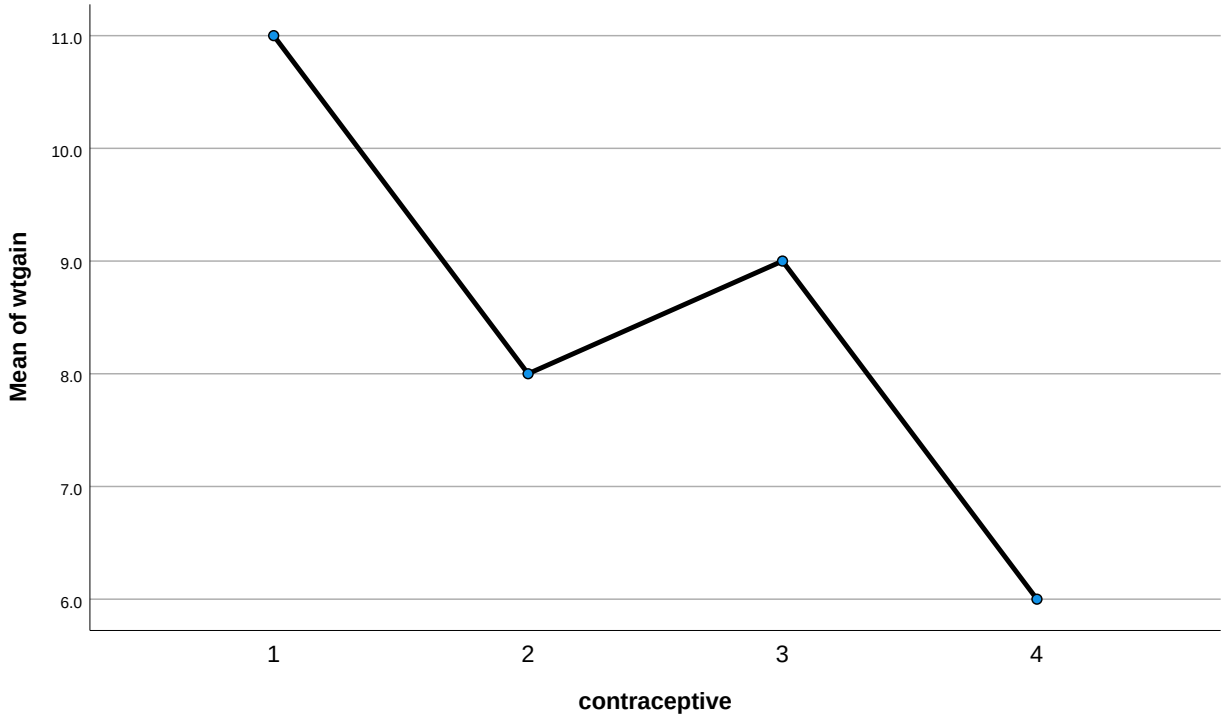
Output Created		03-NOV-2026 15:37:25
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	24
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each analysis are based on cases with no missing data for any variable in the analysis.
Syntax		ONEWAY wtgain BY contraceptive /PLOT MEANS /MISSING ANALYSIS /CRITERIA=CILEVEL (0.95).
Resources	Processor Time	00:00:00.94
	Elapsed Time	00:00:01.02

ANOVA

wtgain

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	78.000	3	26.000	5.164	.008
Within Groups	100.700	20	5.035		
Total	178.700	23			

Means Plots




```
1777  0
1778  0  *------------------------------------------------------------------------------
-----------------------------------------
           ------------------------------------------------------------------------------
---
1779  0
1780  0
1781  0  * End of INSERT and INCLUDE nesting level 02.
1781  0
1782  0  OUTPUT EXPORT /CONTENTS EXPORT=ALL /PDF DOCUMENTFILE='E:/Documents/GitHub/biosta
tistics-portfolio/group-comparisons/spss/sp
           ss-files/contraceptives-one-way-anova.pdf'.
```